

Quantum Complexity of Signal Processing on Higher-Order Networks with Boundary Conditions

Keywords: Topological signal processing, higher-order network analysis, relative Hodge Laplacian, quantum algorithms, quantum complexity

Extended Abstract

Motivation. Recent work in network modeling and analysis has increasingly focused on systems with multiway interactions [1–3], commonly referred to as higher-order networks (HONs). A central challenge in HON analysis is to process and infer signals supported on such structures. Topological signal processing (TSP) [4, 5], which generalizes graph signal processing [6, 7] from graphs to simplicial complexes, provides a natural framework for filtering signals supported on simplices of different orders.

Despite its usefulness for low-dimensional simplicial data, TSP becomes computationally challenging when the underlying network is dense and the signals live on higher-dimensional simplices. The cost grows combinatorially with both the number of nodes and the simplex dimension. Quantum computation offers a possible route to reducing this cost. Although quantum algorithms for TSP have recently been proposed [8, 9], a rigorous complexity-theoretic characterization of their advantage is still missing. Establishing such a characterization would provide a robust framework for the computational complexity of TSP and clarify the extent to which quantum computation can benefit HON analysis.

To study the computational complexity of practically relevant HON signal processing, we focus on a **local boundary-aware inference** problem rather than on full-state filtering. Specifically, we consider the problem of inferring localized signals via TSP filtering under boundary constraints. Such a setting arises naturally, for example, in power or sensor networks, where signals may be grounded or unobserved in certain regions and the goal is to infer signal behavior on an active region from local measurements.

Problem Statement. Let $(\mathcal{K}, \mathcal{L})$ be a simplicial pair, where a subcomplex $\mathcal{L} \subseteq \mathcal{K}$. Then, one can form the *relative* cochain space $C^k(\mathcal{K}, \mathcal{L})$ [10] as a signal space consisting of signals on $\mathcal{K}$, that vanish on $\mathcal{L}$. Also, denote $L_{k,\text{rel}}(\mathcal{K}; \mathcal{L}) : C^k(\mathcal{K}, \mathcal{L}) \rightarrow C^k(\mathcal{K}, \mathcal{L})$ the k -th *relative* Hodge Laplacian [11].

Task 1. Given an input signal $s^k \in C^k(\mathcal{K}, \mathcal{L})$, a probe supported on a simplex or localized region u^k , and polynomial filter p , the task is to estimate either the local filtered response $\langle u^k, p(L_{k,\text{rel}}(\mathcal{K}; \mathcal{L})) s^k \rangle$ or the local filtered energy $\langle s^k, p(L_{k,\text{rel}}(\mathcal{K}; \mathcal{L})) s^k \rangle$.

Note that, in this setting, the subcomplex $\mathcal{L}$ represents grounded, unobserved, clamped, masked, or otherwise inactive parts of the network, so filtering via $p(L_{k,\text{rel}}(\mathcal{K}; \mathcal{L}))$ is performed on the active space and therefore respects boundary constraints by construction. This yields a boundary-aware extension of TSP, which we call *relative topological signal processing* (RTSP).

So, our main question is

“How hard is to solve Task 1 via RTSP filtering on a quantum computer?”

Approach and Methodology. We realize a family of sparse stiffness matrices within relative Hodge Laplacians through (input) oracle-preserving reductions. These stiffness matrices have previously been used to analyze the quantum complexity of simulating classical oscillator networks [12]. Our reduction therefore transfers the corresponding quantum-complexity guarantees for local observables in oscillator networks to natural RTSP filtering tasks. We also formulate a decision version of Task 1 and show, by reduction from the problem in [13], that this decision problem is BQP-complete.

Results. We obtain a rigorous quantum-complexity characterization of Task 1.

Theorem 1 (informal). *Under a sparse-access input model [14], there exists an instance of Task 1 via RTSP filtering that can be solved by a quantum algorithm using polynomially many queries, whereas any classical algorithm requires exponentially many queries to solve the same problem.*

Thus, RTSP is not only a boundary-aware framework for HON signal processing, but also a setting in which provable quantum advantage can be established for natural filtering tasks. Moreover, the corresponding decision version of the task is BQP-complete, showing that boundary-aware HON signal processing is expressive enough to capture universal quantum computation.

Conclusions and Outlook. Relative topological signal processing (RTSP) provides a boundary-aware framework for higher-order signal inference in the presence of grounded, clamped, masked, or unobserved regions. Our results place this local RTSP inference task within a rigorous quantum-complexity framework under sparse-access input models. In particular, we obtain an *exponential quantum-classical query separation* for estimating local filtered responses or energies, while a corresponding decision version is BQP-complete.

By the time of the conference, we aim to complement the theory with (i) numerical case studies illustrating boundary effects in local filtered-response estimation on representative higher-order datasets, and (ii) resource estimates for the quantum routines to identify regimes in which a practical advantage could persist. Key open directions include extending these guarantees to other higher-order operator families and identifying settings in which the advantage reflects intrinsically higher-order structure rather than merely an embedded graph instance. We also note that, for arbitrary sparse matrices, a recent quantum-complexity characterization of matrix functions has been established [15]. Our result further supports the existence of hard subfamilies within sparse matrix-function problems.

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